

Eashwar Siddha

AI & Computer Science Undergraduate | Machine Learning Engineer | Ag-Tech Enthusiast

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PROFESSIONAL SUMMARY

Results-driven AI and Computer Science undergraduate with hands-on experience in machine learning, deep learning, computer vision, and full-stack development. Proven track record delivering production-grade AI/ML solutions in agriculture technology, cybersecurity, and biomedical imaging. Skilled in Python, TensorFlow, PyTorch, and cloud-integrated pipelines. First Class Honours in Year 1 at the University of Nottingham Malaysia.

EDUCATION

University of Nottingham, Malaysia — BSc (Hons) Computer Science with Artificial Intelligence

September 2024 – Present

- Year 2 Undergraduate — First Class Honours in Year 1 (top academic standing).
- Head of Technology, Computer Science Society (Academic Year 2026–2027): leading digital initiatives, managing technical infrastructure, and driving tech-focused events and workshops.
- Notable member of the Computer Science Society, Publicity Team.

The Velammal International School, Chennai — High School Diploma

June 2022 – May 2024

- Mathematics and Computer Science major with a 93.2% weighted average.

EXPERIENCE

University of Nottingham, Malaysia — Student Research Intern

July 2025 – September 2025

- Designed and implemented a U-Net based semantic segmentation pipeline for automated seed analysis, enabling precise radicle and plumule measurement to support quality assurance workflows in seed distribution companies.
- Applied deep learning and image processing techniques to achieve high-accuracy segmentation on microscope slide datasets.

Dr. Ken Kahn, University of Oxford (Remote) — Student Intern

August 2023 – October 2023

- Collaborated with Dr. Ken Kahn (Retired Lecturer, University of Oxford) to develop a VGG16-based real-time rotten fruit detection system using convolutional neural networks and transfer learning.
- Delivered a functional computer vision solution demonstrating strong proficiency in image classification and model fine-tuning.

PROJECTS & RESEARCH

Farmer Buddy — All-in-One Precision Agriculture Mobile App — Agritix Sdn Bhd

- Developed in collaboration with Agritix Sdn Bhd: a full-stack mobile application for farmers integrating an AI-powered Agronomist Chatbot, IoT data visualization, plant disease detection, irrigation alerts, scheduling, and body-cam employee monitoring.
- First Place — Best School Project Category.

Metaheuristic Algorithm Benchmarking for IDS Classification — Research Study

- Conducted a comparative study evaluating Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), Hybrid GA-PSO, and NSGA-II for feature selection on an Intrusion Detection System (IDS) classifier.
- Analyzed trade-offs between classification accuracy, feature reduction rate, and computational efficiency across all five optimization strategies.

PMY Segmentation Pipeline — Image Processing Research

- Engineered an end-to-end image processing pipeline for Polymorphonuclear Yeast (PMY) cell segmentation from microscope slide images.
- Applied classical and deep learning-based computer vision techniques to isolate and quantify cellular structures for biomedical analysis.

Seed Germination Analyzer — U-Net Semantic Segmentation

- Built a U-Net architecture model to perform pixel-level semantic segmentation and quantify radicle and plumule emergence in seed quality assurance applications.

Aegis WAF — *NeMo Guardrails + CaMeL Architecture*

- Developed a Web Application Firewall security tool for FinTech agents using a two-tier CaMeL (Carefully Managed LLM) architecture with dynamic risk scoring to prevent prompt injection attacks.

OCR Handwriting Recognition — *CRNN Deep Learning Model*

- Replicated the keras_ocr library by training a custom CNN-CRNN architecture on the IAM Handwriting Dataset, achieving 76% accuracy on handwritten text recognition.

Pocket Farmer — *VGG16 Real-Time Rotten Fruit Detector*

- Built a real-time and image-based rotten fruit detection system leveraging VGG16 transfer learning, demonstrating practical expertise in computer vision and AI/ML model deployment.

WebCam Paint — *MediaPipe Gesture Recognition App*

- Developed a fully functional real-time paint application using MediaPipe hand tracking, where the webcam serves as the canvas and finger gestures act as paintbrushes.

Maze Solver — *Search Algorithm Visualizer*

- Implemented an interactive maze-solving visualizer supporting BFS, DFS, A*, and other search algorithms, enabling real-time comparison of pathfinding strategies.

SRTF Visualizer — *Process Scheduling Algorithm CLI*

- Created a command-line visualizer for the Shortest Remaining Time First (SRTF) CPU scheduling algorithm, simulating process queues and Gantt chart output.

Contact Management System — *CLI Tool — University Coursework*

- Designed and built a robust command-line contact management application with CRUD operations, search, and data persistence as part of university coursework.

Tetris JavaFX Clone — *Code Refactoring & Feature Engineering*

- Refactored and fully revamped a buggy JavaFX Tetris clone, resolving critical logic errors and adding new gameplay features using object-oriented design principles.

TECHNICAL SKILLS

Programming Languages: Python, C, Java, Haskell, ARM Assembly, HTML, CSS, PHP, React.js, TypeScript, JavaScript

AI / ML Frameworks: TensorFlow, PyTorch, OpenCV, Keras, MediaPipe, YOLO, U-Net, Mask R-CNN

Research & Algorithms: Genetic Algorithm (GA), PSO, ACO, NSGA-II, Hybrid GA-PSO, Feature Selection, Multi-Objective Optimization

Other Tools: Hardware Simulation / Verilog, Git, LibGDX, JavaFX, Caddy, LiveKit, NeMo Guardrails

Key Competencies: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), Image Segmentation, Transfer Learning, Full-Stack Development, Agile Development

AWARDS & ACHIEVEMENTS

- First Place — Best School Project Category, Farmer Buddy (Agritix Sdn Bhd collaboration).
- SAT India Top Performer Award 2023.
- India Top 96 Semi-Finalist — TCS InQuizitive National Quiz Competition.

LANGUAGES

English (Fluent) | Tamil (Native) | Saurashtra (Native)

INTERESTS

Music & Guitar | Web & App Development | Community Volunteering | Hiking & Camping